

Cell Division

Mitosis produces two genetically identical daughter cells from a single parent cell. The process is divided into prophase, metaphase, anaphase, and telophase. Mitosis is responsible for growth and tissue repair in multicellular organisms.

Meiosis produces four genetically distinct gametes from a single parent cell. Crossing over during prophase one creates new combinations of alleles. The result is increased genetic variation in sexually reproducing populations.